

Lab 1 – Data & Information

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1 Data vs Information

Data is simply "stuff" in the world. Data can take the form of anything... numbers, lines, or words. Data, however, cannot make conclusions. Information is data empowered with the context to make useful statements the trends or lack there of. For example, if we had the numbers 98 90 70 63 they are meaningless without context; they carry no information, but they are still perfectly valid data. Now let's add some context: those numbers are 4 grades from a professors class of 20. Now this data means something, it represents some information. We could make statements such as the average of these grades are 80.25 because it makes to average these numbers in the given context. We could even try to make farther reaching statement such as the average for the entire class is probably around 80.25. Still, we must be careful with how much information we actually have and whether can accurately make certain assumptions. In example, the context and information could change: let's say theses were the top 4 student grades. The data remains the same but the context changes everything. There a myriad of conclusions you might want to make about this data but have you considered all the information and especially the missing information? What is the grading scale? Are these even final grades? What is rigor of the class?

Databases attempt to organize data in efficient ways whose structure, most importantly, aims to convey all of the respective information. Organizations who maintain a lot of data want to embed their "business rules" directly into how they store their data, and database are perfect for this. For example, let's look at the Marist Banner system. (I not have access to the actual database so this is inferred from the api their frontend website uses, an example section json entry is below for reference) Firstly, Banner stores this "id" attribute. From the

outside this seems like a completely obscure number with no meaning. And, it is, this number does not hold any meaning in the way numbers generally have meaning. It is likely only there to give each section a unique identifier. Without being coupled with the information of being an "id" this data is a completely useless number. Sections also likely have a one-to-many relationship with "faculty" and "meetingsFaculty". This relational idea gives information: there is a group of "faculty" and "meetingsFaculty" which which can be apart of a section. Which, from what we know about courses at Marist, makes sense. Every section has one to many teachers and some meeting times. Databases organize data in order applying the business logic of the organization forming their data into information.

2 Data Models

Hierarchical databases store their data through a tree-like structure conveying information through relationships such as parent and child. Every child can only have a single parent and a parent can have an indefinite of children. These constraints nicely represent data that can form perfect trees. However it breaks down when any other relationship needs to be represented. For instance, let's go back to the Marist database imagine a simplistic version of storing sections, meetings, and their teachers. In a hierarchical database sections and meetings are a perfect match. At least simplistically, every section has 0 to many meetings and every meeting has exactly one parent. However, representing their teachers is difficult even if we simplify that each section only has a single teacher. Are they children of section? Then we would lose the information that children of sections are meetings. Are they children of meetings? This could work but then we would be storing professor data for each meeting. Whenever data is repeatedly stored it not only takes up more space but even more worryingly can lead to update, insertion, and deletion anomalies.

Maybe another type of data models can help us! Network databases permit a child to have more than one parent. This means we can have one instance of professor and all of their meetings point to them. We maintain the structure/information that children of sections are meetings as well as children of meetings are professors without duplication. This is great until we have a professor that does not teach any meetings. Where should we put this professor? There is no good answer in either of these databases. This is one of the fundamental reasons why relational databases are largely thought as better.

There is another type of storage method known as XML. I personally have to appreciate XML as I use HTMX, a small javascript library which uses html attributes to send AJAX requests to an endpoint. I can represent all of my needs to send information from my frontend to my backend. However, I think it is better suited as a data transfer structure rather than a full data storage and query system because it runs into the same problems as network database.

XML does allow multiple parents and children but still does not have a good place for data that is not connected.

3 Proof of Life

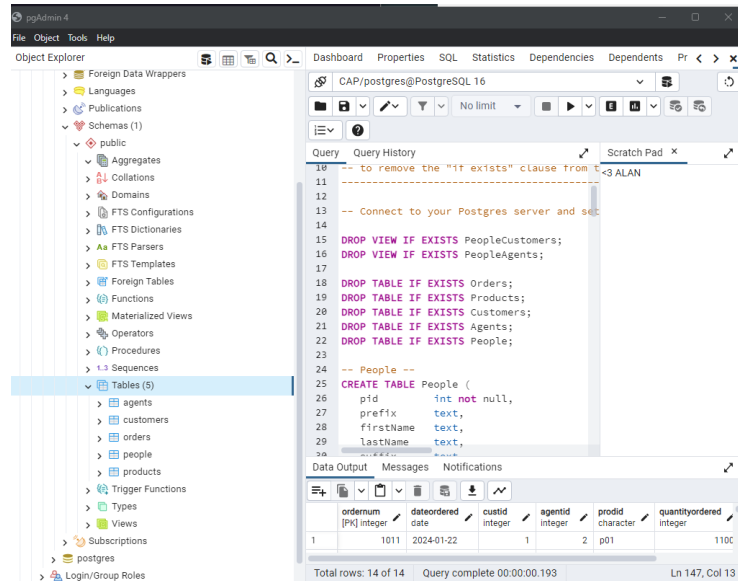


Figure 1: PG admin running on my computer showing the cap database

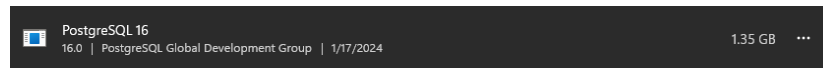


Figure 2: Postgre sql installed

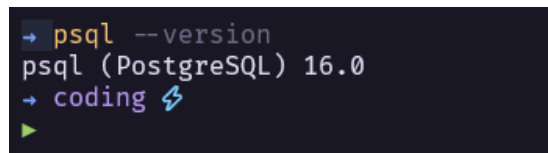


Figure 3: psql cli tool working in powershell

```

1 {
2   "id": 164935,
3   "term": "202320",
4   "termDesc": "Spring 2023",
5   "courseReferenceNumber": "12114",
6   "partOfTerm": "1",
7   "courseNumber": "331L",
8   "subject": "CMPT",
9   "subjectDescription": "
10    Computing Technology",
11   "sequenceNumber": "111",
12   "campusDescription": "Marist
13    College Campus",
14   "scheduleTypeDescription": "
15    Lecture",
16   "courseTitle": "THEORY OF
17    PROGRAM LANG",
18   "maximumEnrollment": 25,
19   "enrollment": 22,
20   "seatsAvailable": 3,
21   "waitCapacity": 0,
22   "waitAvailable": 0,
23   "creditHourLow": 3,
24   "openSection": true,
25   "faculty": [
26     {
27       "bannerId": "38372",
28       "displayName": "Labouseur,
29       Alan",
30       "emailAddress": "Alan.
31       Labouseur@marist.edu",
32       "primaryIndicator": true
33     }
34   ],
35   "meetingsFaculty": [
36     {
37       "category": "01",
38       "meetingTime": {
39         "beginTime": "0800",
40         "building": "HC",
41         "buildingDescription": "
42         Hancock",
43         "campus": "AAA",
44         "campusDescription": "Marist
45         College Campus",
46         "category": "01",
47         "creditHourSession": 0.0,
48         "endDate": "05/12/2023",
49         "endTime": "1215",
50         "friday": true,
51         "hoursWeek": 1.25,
52         "meetingType": "LEC",
53         "meetingTypeDescription": "
54         Lecture",
55         "monday": false,
56         "room": "2005",
57         "saturday": false,
58         "startDate": "01/17/2023",
59         "sunday": false,
60         "thursday": false,
61         "tuesday": false,
62         "wednesday": true
63       }
64     },
65     {
66       "category": "01",
67       "meetingTime": {
68         "beginTime": "1100",
69         "building": "HC",
70         "buildingDescription": "
71         Hancock",
72         "campus": "AAA",
73         "campusDescription": "Marist
74         College Campus",
75         "category": "01",
76         "creditHourSession": 0.0,
77         "endDate": "05/12/2023",
78         "endTime": "1215",
79         "friday": true,
80         "hoursWeek": 1.25,
81         "meetingType": "LEC",
82         "meetingTypeDescription": "
83         Lecture",
84         "monday": false,
85         "room": "2005",
86         "saturday": false,
87         "startDate": "01/17/2023",
88         "sunday": false,
89         "thursday": false,
90         "tuesday": false,
91         "wednesday": false
92       }
93     }
94   ],
95   "sectionAttributes": []
96 }

```

Listing 1: A Course's section from Banner's api